Paper 8

Mass as the Internal Accounting of Departure from c

Paper 8 of the Series

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ABSTRACT

Paper 6 established that a massive configuration runs a ceaseless internal scan and that its processing budget is split between translation and that scan, and Paper 7 carried the split forward and named two debts as belonging to this paper: the quantitative law of mass increase toward the photon baseline rate, and the limiting behaviour that forbids a massive configuration from ever reaching it. Paper 8 pays both debts, and it does so from a single mechanical fact established in Paper 4: mass is not a substance a configuration contains but the running cost of holding a tilt away from the photon baseline. A photon, State 9, holds no tilt; it runs only an identity scan, pays the fixed minimum identity skim that any configuration owes simply to be itself — three percent of the total budget, the Circa — and commits the whole remaining ninety-seven percent to translation. That ninety-seven percent of Circa is the speed of light c: a fixed universal constant, the same for every configuration, being the largest share of the one budget that can ever reach motion once existence is paid for. The first departure from that baseline is a single-axis tilt — State 7 or State 8, the two configurations that mark the first resolved step away from the photon and fix, by which of the two it is, the direction of that departure across the framework's axes. The instant a configuration holds such a tilt it can no longer run an identity scan alone; it must also run a temporal scan to track its route of resolution back toward baseline, and the union of the two is the manifold scan. Mass is the cost of running the manifold scan rather than the bare identity scan — the share of the fixed budget diverted from translation into holding the tilt. From this one accounting the paper derives what the standard account postulates. It shows that mass and energy are the same geometric fact, energy being the work to cut a tilt and mass the standing ledger that holds it, and it derives the equivalence $E = m \cdot c^2$ as a topological displacement cost, with c^2 entering not as an arbitrary scaling constant but as the geometric area of the processing boundary across which a standing loop must converse with its neighbours, squared because the identity and temporal sweeps of the manifold scan each scale with the boundary rate c. It then derives the quantitative law Paper 7 assigned here: because the ninety-seven percent that remains after the identity skim splits between spatial translation and the temporal scan, and the two are orthogonal demands on one conserved total — c, the fixed translation ceiling — they stand in a right-angled relation, and the closed form that drops out of that relation is exactly the Lorentz factor, the square root of one minus v-squared over c-squared. The same accounting shows why a tilted configuration can never reach c: doing so would demand committing the whole ninety-seven percent to translation, driving the temporal scan to zero, at which point the tilt can no longer be held and the configuration dissolves back into the baseline. The budget rule is shown to extend isotropically across the three native axes, so that the internal scan rate throttles with total speed alone and never with direction, and the escalating workload near c is identified mechanically as a lengthening route of resolution computed on a starving budget. The threshold where mass switches on is mapped in full: from the zero of State 9, through the single-axis first tilt of State 7 or State 8 that fixes the minimum mass quantum at the resolution floor, to the three-rotation cubic lock of State 1. The same accounting — a fixed identity skim taken off the top of Circa, then a translation-versus-temporal-scan split of the ninety-seven percent that remains — yields the velocity hierarchy in full, from the photon that runs no temporal scan and travels at c to the singularity whose skim consumes everything and which cannot move at all. Throughout, mass is expressed in the framework's single ruler — departure from the photon baseline, measured against the Discrete-Step Resolution Floor of Paper 4 — and never in kilograms, the kilogram being the arbitrary human artifact this series exists to replace. The gravitational reading of the same

departure is deferred to Paper 10, the time-dilation that the budget split produces to Paper 13, and the composite mass of bonded matter to Paper 16.

1. The Question Mass Actually Asks

The standard account treats mass as a primitive. A body *has* a mass, a number measured against a metal prototype and written in kilograms, and that number enters the equations of motion as a given: it sets how hard the body is to accelerate, how strongly it gravitates, and — through the most famous equation in physics — how much energy the body is equivalent to. The number is measured, tabulated, and used, but its origin is left unexplained. Why does a body resist acceleration at all? Why should that resistance be the very same quantity that gravitates? Why should it convert to energy, and at the enormous exchange rate set by the square of the speed of light? The standard account answers none of these mechanically. It postulates the relations and confirms them to extraordinary precision, which is a different thing from explaining them, and it measures the quantity in a unit — the kilogram — that has no connection to anything physical except a chosen artifact, which is precisely the fragmented metrology Paper 4.5 audited.

This paper asks what mass *is*, in the sense the series has asked of every quantity it has treated: what is the configuration actually doing, on the medium, such that we observe the thing called mass? The answer was seeded in Paper 4, which established the photon as the baseline and the electron-positron pair as a tilt away from it, and in Papers 6 and 6.5, which established that a departed configuration runs a ceaseless scan and pays for it out of a fixed budget. This paper develops those seeds into a quantitative law. Mass is not a stuff. It is the running cost of a commitment — the cost of holding a tilt away from the photon baseline, paid every tick out of the one budget every configuration has to spend.

The standard of evaluation is the one fixed in Paper 1 and held throughout: complete and transparent accounting, every quantity built from what the framework has already established, no number reached for that the framework has not earned, and one ruler only. By the end of this paper, mass must be expressed entirely as departure from the photon baseline, measured against the resolution floor of Paper 4 — not in kilograms, and not as a primitive — or the paper has failed its own standard.

Two boundaries of scope are declared here at the outset, so that neither can be mistaken for an omission. First, on composite mass: the functional derivations for composite, multi-junction material mass — atomic clustering, multi-body binding overhead — are not treated in this paper, and their absence is deliberate architecture, not a gap. A common failure of continuous formalisms is the monolith compendium, the thousand-page volume that buries its fundamental mechanics beneath endless composite equations; this series refuses that style in favor of a modular blueprint. Paper 8 is dedicated strictly to the internal accounting of an individual configuration's tilt away from the photon baseline. To inject the network overhead of multi-body locks into these baseline proofs would clutter the ledger; that accounting is partitioned to its own coordinate, Paper 16 (composite and junction mass). Second, on units: the omission of the kilogram and every other dimensioned human unit is likewise a deliberate architectural constraint, not a limitation. The kilogram is a localized artifact of terrestrial engineering, absent from the universal ledger; to inject a human-scale metric constant into these proofs would pollute a mass accounting that is, by design, a pure dimensionless ratio of departure from the photon baseline. The conversion bridges that would translate these ledger states into human measurements are a separate downstream task, and they change none of the internal ratios derived here.

2. The One Budget and the Two Scans

The single load-bearing commitment is the monolithic processing rate established across Papers 2, 4, and 6 and reaffirmed in Paper 7: every configuration in the medium is allotted exactly one total processing budget per tick, the **Circa**. This is the hardware clock of the medium itself — the full sovereign pool of processing capacity, identical for a

photon and for a planet. No configuration is given more, and none less; what differs between a photon and a massive thing is not the size of the Circa but how it is spent. The speed of light, as this section will show, is not the Circa itself but a fixed fraction of it — the largest share any configuration can ever route into motion once the unavoidable cost of existing has been paid.

Existence is skimmed off the top of Circa first. Before a configuration can spend anything on motion, it must first establish that it exists — it must run its **identity scan**, the structural loop that confirms what it is across the three native axes (Paper 6). That scan is never free: a configuration that ran no identity scan would not know it was anything, would log no entry on the ledger, and would not exist at all. The identity scan is therefore a skim taken off the top of Circa, before any motion is budgeted, because energy cannot be manufactured from nothing and existence must be paid for first. The minimum skim, paid by the simplest possible configuration, is three percent of Circa. This figure is not fitted; it is the framework's own three expressing itself in the budget — the same three that gives the medium its three axes, its three per-axis values (+1, 0, −1), the three sides of the resource triangle of Section 5, and the three rotations that build a State 1. No configuration pays less than this three-percent floor, because none can exist while confirming less than its bare identity.

The speed of light is the ninety-seven percent that remains — a fixed universal constant. What is left of Circa after the minimum identity skim is ninety-seven percent, and that is the most any configuration can ever commit to spatial translation. **That maximum is what the speed of light, c, is.** The speed of light is not the whole Circa, and it is not an object flying through empty space at full processing capacity; it is the velocity of a configuration that spends nothing on tracking state-changes, leaving the largest possible remainder — ninety-seven percent of Circa — for spatial coordinate updates. Read this way, c is a fixed universal constant: ninety-seven percent of the one Circa, the same number for every configuration in the medium, because Circa is the same for all and the minimum identity skim is the same for all. This is the point on which the whole accounting must be exact, and it answers a real objection directly: the speed limit does not slide from particle to particle. A heavier configuration does not get a smaller c; it gets the same c as everything else, and what changes is only how much of that ceiling it can actually use for motion once its temporal scan has taken its share. The ceiling is universal; the usage varies.

The photon reaches c because it runs no temporal scan. A configuration at State 9 holds a central-junction signature of pure zero — no axis tilted into the centre. It pays the minimum three-percent identity skim, like everything else, simply to register its presence. But because it holds no tilt, it has nothing to track: it never changes state, so it runs **no temporal scan at all**, and its temporal scan cost is exactly zero. This is the precise content of the photon being "massless" — not that its total workload is zero, but that its *temporal* scan is zero, so it dedicates not one cycle to axis-tilt tracking or configuration duration. With zero temporal scan, the photon routes its whole remaining ninety-seven percent of Circa into translation, and so it travels at exactly c. And running no temporal scan, it tracks no duration of its own — which is why light experiences zero proper time, a fact the framework recovers rather than postulates.

The first tilt switches on the temporal scan. The first departure from the baseline is not the electron — it is a *single-axis* tilt, State 7 or State 8, the two configurations in which exactly one axis has stepped a non-zero value into the centre while the other two remain at zero. These two are the first resolved marks away from the photon, and which of the two a configuration is fixes the *direction* of its departure: State 7 and State 8 are the opposite single-axis steps, and the choice between them is what later determines whether the configuration is built up X-dominant or Y-dominant as further rotations are added. A full State 1 configuration — the electron or positron — is not the first tilt; it is the fully locked cubic mark reached only after **three rotations** carry the single-axis first step up to a saturated three-axis lock. Mass switches on at the first of those rotations, at State 7 or State 8, not at State 1.

The instant a configuration holds a tilt, it has something to track, and it must run a **temporal scan** on top of its identity skim — the scan that follows the configuration's route of resolution back toward the baseline it departed from. The identity scan asks *what am I*; the temporal scan asks *how far am I from home and by what route do I*

return. The temporal scan is drawn from the ninety-seven percent that would otherwise translate the configuration: every cycle it claims is a cycle no longer available for motion. The union of the identity scan and the temporal scan is the **manifold scan**, the full scan every massive configuration runs, established as the partner-query architecture in Paper 6 and named here in its two parts. It is the temporal scan's bite into the ninety-seven percent — the budget stolen from translation to track a tilt — that registers as mass.

This is the origin of mass, stated mechanically:

Mass is the cost of the manifold scan that a tilt forces — the identity scan grown above its three-percent floor to confirm a complex structure, plus the temporal scan that structure triggers to track its resolution — all of it budget drawn from Circa and no longer available for translation.

A photon pays the three-percent identity floor, runs no temporal scan, and commits its whole ninety-seven percent to translation; it is massless and moves at c. A tilted configuration runs a heavier identity scan — more structure to confirm — which triggers a temporal scan to track that structure's route home, and both draw from Circa, so less of the ninety-seven percent reaches translation. That shortfall against the photon's maximal motion, measured in the single ruler, is the configuration's mass. The identity scan is the cause and the temporal scan the effect: a more complex structural identity takes more cycles to verify, and that heavier identity natively triggers a heavier temporal calculation to track its resolution, so the two climb together. This is why mass rises with the magnitude and complexity of the tilt, from the single-axis first step of State 7/8, through the three-rotation climb, to the saturated cubic lock of State 1 and beyond to the dense multi-junction tilts of composite matter.

The boundary accounting of the baseline photon. Because the photon is the zero of the entire ruler, its budget deserves to be stated to the last percentage point, and it exposes a distinction the standard account never draws. Standard physics equates "massless" with zero internal processing — a state of pure translation at a hundred percent, with nothing spent internally. This framework rejects that as an unmetered free-existence, a something that pays nothing to be itself. Every configuration the medium sustains must run an identity scan to hold its coordinate existence, and that scan is a fixed, non-negotiable three percent of Circa per tick, drawn off the top because energy cannot be manufactured from nothing. The photon pays that three percent like everything else. What makes it massless is not that its total workload is zero but that its *temporal* scan is exactly zero: holding no tilt, it has no state-change to track, and it dedicates not one cycle to axis-tilt tracking or duration. Its remaining ninety-seven percent is therefore committed wholly to translation, and that ninety-seven percent of Circa is the velocity the macroscopic observer records as c. When a configuration departs the baseline, its temporal scan rises above zero, those cycles are subtracted from the ninety-seven percent, and its translation velocity is forced below c. The photon thus sits at the exact corner of the ledger where identity is paid in full, temporal scan is zero, and translation is maximal — the one configuration that spends nothing on time and therefore everything left on space.

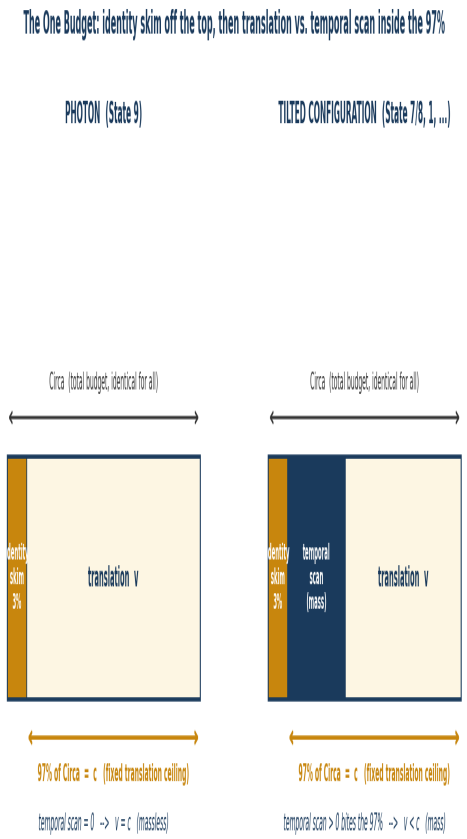


Figure 1 - The Budget of Circa. The three-percent identity skim comes off the top of the one universal budget; the remaining ninety-seven percent is the fixed translation ceiling c. The photon (left) runs zero temporal scan and commits the whole ninety-seven percent to translation; a tilted configuration (right) runs a temporal scan that bites into the ninety-seven percent, and that bite is mass.

3. The Sum Rule and the Two-Column Ledger

The circa is always fully spent. Nothing is held in reserve and nothing leaks, under the Strict Generative Accounting locked in Paper 7. For a tilted configuration the spending falls into two columns, and they always sum to the same fixed total:

(budget in translation, v) + (budget in the temporal scan, s) = c, the fixed translation ceiling, always.

This sum rule governs the ninety-seven percent of Circa that remains after the identity skim of Section 2 has been paid. Its ceiling, c, is the same fixed constant for every configuration — ninety-seven percent of the one universal Circa — not a quantity that shrinks for heavier configurations. The rule is exact at every speed, because it is only the statement that this ninety-seven percent is always fully spent, split between translation and the temporal scan. What changes as a configuration speeds up or slows down is the *split* between the two columns. A configuration barely moving spends little of the ceiling on translation and most of it holding its tilt; a configuration driven hard toward c spends more and more on translation, leaving less for the temporal scan that holds it together. The identity skim sits outside this rule, already taken off the top of Circa; the sum rule is what the configuration does with what the skim leaves behind.

Two limiting reads frame everything that follows. At one limit sits the photon: the minimum three-percent skim, no tilt, no temporal scan, so its whole ninety-seven percent goes to translation — s is zero, v equals c, mass is zero. At the

other limit sits the impossibility for a tilted configuration: driven so hard that translation claims the entire ceiling, leaving nothing for the temporal scan that holds its tilt. That limit cannot be reached, because a configuration that cannot run its temporal scan cannot hold its tilt, and a tilt that cannot be held dissolves back into the baseline. The approach to c is therefore an approach to self-dissolution, and the cost of the last increment toward it is what the next two sections quantify — first as the energy locked in the tilt (Section 4), then as the law of how that cost escalates toward c (Section 5).

4. Mass and Energy as One Geometric Fact: The Derivation of $E = m \cdot c^2$

The framework treats energy and mass not as two quantities linked by a conversion but as one geometric fact read in two states. This section derives their equivalence as a strict displacement cost, in ledger steps, using only what the series has already fixed.

Step 1 — the zero reference. Begin at State 9, the photon. Its central-junction signature is pure zero; it holds zero displacement from the sovereign zero; it pays the minimum three-percent identity skim off Circa and, with no tilt and so no temporal scan, commits its whole remaining ninety-seven percent to translation at velocity c. Its standing mass-energy is identically zero. This is the absolute reference against which everything is measured — the single ruler.

Step 2 — the work of the tilt. To make a massive configuration, geometry must be delivered across the configuration's boundary. That delivery is the framework's definition of **energy**: a deterministic injection of topological distortion across a continuous boundary (Paper 4, Paper 6.5). When the injected geometry forces an axis-tilt — the first single-axis step of State 7 or State 8, and onward through the rotations — it deforms the local medium into a standing crease that can no longer slide freely at c. The energy is the total work done to cut that crease into the medium.

Step 3 — the workload of holding it. A continuous medium cannot hold an unreconciled crease passively; the configuration must defend the tilt by running its manifold scan every tick (Section 2). The tilt of a given magnitude therefore carries a fixed, mandatory workload: a specific share of the circa committed, every tick, to the temporal scan that holds the displacement against the Discrete-Step Resolution Floor a-k of Paper 4 — the non-zero minimum interval of the medium below which no crease can be resolved, cited there (Section 3 and Appendix) as the floor that lets a State-1 tilt stand without dissolving. The workload to hold a tilt is the standing energy of that tilt, paid continuously rather than once.

It is critical to establish the strict operational limitation of the resolution floor a-k here, to prevent the coordinate assumptions of continuous field theory from misreading its presence. The marker a-k possesses no metric scale, carries no dimensioned value, and cannot be scaled, expanded, or varied. It is not an empirical constant and not a particle radius. Its single, invariant function is that of a logical delimiter on the processing manifold: it exists only to distinguish one from two, a step from no step, a direction from its opposite — the minimal punctuation the ledger requires to tell a localized configuration apart from the baseline medium and to establish directionality across the three axes. Because every physical law derived in this paper — the Lorentz factor, the equivalence $E = m \cdot c^2$, the mass ordering — depends strictly on dimensionless ratios of how the fixed circa budget is split ($v^2 + s^2 = c^2$), the absolute size of the delimiter is mathematically irrelevant to every result. A reviewer trained on continuous metrics will look for a dimensioned constant and try to take a limit as a-k shrinks to zero, but there is no such limit to take: a-k is a binary rule of structural identity, not a metric parameter, and it cannot be leveraged to smuggle an un-derived scale into the ledger. It is not a measurement; it is the baseline hardware rule that makes a ledger entry possible at all.

Step 4 — why the conversion factor is c-squared. The manifold scan is not one sweep but two: the identity scan and the temporal scan, the double-sweep established as the two-phase resolution check of Paper 2 (the 720-degree

double traversal a tilted configuration requires to return to its own orientation). Each sweep converses across the configuration's processing boundary, and each scales with the boundary rate c, because c is the rate at which the medium processes any conversation across that boundary. The standing loop must hold against *both* sweeps at once, across the full two-dimensional extent of its processing boundary — and the measure of that extent is an area, the product of the two boundary sweeps, each carrying a factor of c. The conversion factor between a quantity of held tilt and the energy required to hold it is therefore not an arbitrary constant but the **geometric area of the processing boundary**, $c \cdot c = c^2$. The square is the signature of the double-sweep: one factor of c for the identity sweep, one for the temporal sweep, multiplied because the scan must satisfy both across the boundary simultaneously.

A reviewer will object that a 720-degree traversal is two 360-degree loops in sequence, so the cost should *add* — c + c = 2c — not multiply. The objection mistakes the mechanism, and the answer is that the temporal sweep is not a second, independent loop run after the first; it is a mirror of the identity sweep, run at the same time. The identity sweep is the configuration's built-in confirmation of *what it is* — already running natively, the baseline existence loop on the three-percent floor. The temporal sweep takes that very identity and reflects it: knowing what it is, it traces the route that identity must follow back to the photon baseline. The temporal sweep is therefore not new information computed separately; it is the identity reflected into the question of resolution, and because it is a reflection of the identity it must run on the same budget, across the same boundary, on the same tick. The two cannot alternate — the configuration cannot pause confirming what it is in order to compute its way home, because a tilt left unconfirmed for even one tick dissolves (Section 2). They are locked simultaneous: identity along one facet of the boundary, its mirror along the orthogonal facet, both satisfied at once. When a value is projected against its own mirror simultaneously, the two measures multiply into an area rather than adding into a length. The two sweeps are not two separate processes whose costs add; they are two orthogonal facets of one boundary that must both be satisfied, and a boundary with two orthogonal facets each of length c has area c^2 — this is the geometric content of the double-sweep, not an algebraic composition rule for two separate energies. That is why the cost is c × c and not c + c: the 720-degree geometry makes the two sweeps mandatory on every tick, and the mirror relation makes them simultaneous rather than sequential, so their measures form the square that is the processing-boundary area. The square is not forced to fit $E = m \cdot c^2$; it is what a simultaneous reflection across a two-faceted boundary necessarily is.

Step 5 — the substitution. Define **mass** strictly as the standing ledger entry that tracks the held displacement: the magnitude of tilt a configuration carries, measured against the baseline. Call that ledger entry m. The energy delivered to cut and then hold that tilt is the work done across the processing boundary, and that work is the tilt-magnitude multiplied by the area of the boundary across which it must be held:

$$\mathbf{E = m \cdot c^2}$$

Energy is the tilt read as the work to cut and hold it; mass is the same tilt read as the standing ledger that holds it; and c^2 is the area of the processing boundary that converts between the two. They are not two things related by a formula — they are one displacement, accounted two ways, with c^2 the geometric exchange rate fixed by the double-sweep of the manifold scan. No kilogram appears; m is a count of tilt-magnitude against the photon baseline, and c is the medium's own rate.

Step 6 — annihilation as the exact mechanical inverse. The equivalence is not abstract; it describes electron-positron annihilation directly. A State-1 electron and a State-1 positron are the same configuration tilted in opposite directions (Paper 4). When they share a junction, their opposing creases align, and because the opposite tilts cancel, the net tilt drops instantly to zero. With zero tilt there is nothing for a temporal scan to hold; the manifold scan collapses back to a bare identity scan; the standing ledger entry m drops to zero. The geometry that was locked in the tilt — the held energy $m \cdot c^2$ — has nowhere to be filed except back into the medium, and it is shed into the adjacent coordinates as an un-tilted wave running at the full baseline rate: photon radiation. The familiar statement that the released radiation carries energy equal to $m \cdot c^2$ is, in this framework, the exact bookkeeping of a

tilt being returned to baseline and its held workload released as translation. Annihilation is topological restoration to State 9, and $E = m \cdot c^2$ is its conservation law.

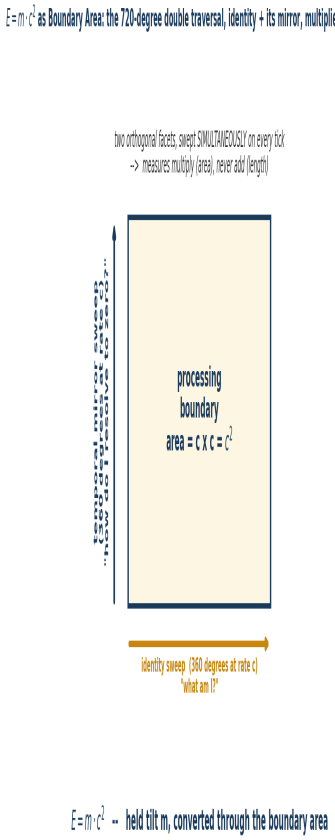


Figure 4 - E = m·c-squared as Boundary Area. The identity sweep and its temporal mirror run simultaneously across orthogonal facets of one processing boundary, each at rate c; their measures multiply into the area c-squared that converts held tilt to energy.

5. The Law of Mass Increase: Deriving the Lorentz Factor

Paper 7 assigned to this paper the quantitative law of why a configuration's ledger workload escalates as it is driven toward c. The budget mechanics of Section 3 yield that law in closed form, and the form is exactly the one relativity postulates.

5.1 The Orthogonal Split and the Pythagorean Constraint

Within the ninety-seven percent of Circa that remains after the identity skim of Section 2 — the budget available for action, whose ceiling is the fixed universal constant c — the capacity splits into two demands: spatial translation, which steps the configuration's directional offset across the discrete intervals of the medium, and the temporal scan, which holds the tilt. These two are *orthogonal* demands on that budget — translation steps the configuration *through* the medium, while the scan holds it *against* the medium — and they draw on a single conserved total, the fixed ceiling c. The right-angled relation between them is not an aesthetic choice made because it happens to yield the Lorentz factor; it is forced by three architectural rules of the medium, and it is worth stating them, because the orthogonality is exactly the seam a reviewer will press.

First, the three-axis hardware: the medium is defined by three native dimensions, X, Y, and Z, and every translation or structural tilt must project across these three headings, with no unmapped fractional dimension and no

background metric to absorb a loose processing cost. Second, the three-value ledger rule: on any single axis the ledger recognizes exactly three states — +1, a step one way; −1, a step the other; 0, flat baseline — the three discrete values the delimiter a-k logs as structural directionality. Third, zero cross-talk: because the ninety-seven percent is a closed budget, translation steps and temporal-scan cycles must be completely independent draws on the one ceiling c. If any part of the translation budget could simultaneously perform the scan, or any part of the scan budget could help step the configuration through space, the ledger would be double-counting cycles — spending the same tick twice — which violates the Strict Generative Accounting locked in Paper 7. Orthogonality is the geometric expression of that zero cross-talk: two draws that cannot lean into or assist one another sit at a right angle of absolute operational divergence, because a component of one along the other would be precisely the double-counting the accounting forbids.

When an independent resource draw is projected across a space bounded by exactly three axes and governed by three directional values, the resulting ledger has no geometric choice but to close into a figure with three sides — a triangle. The translation rate v and the temporal scan rate s are its two perpendicular legs, forced perpendicular by zero cross-talk, and the fixed ceiling c is its hypotenuse. The triangle is not borrowed to make the Pythagorean theorem available; it is the only closed shape a three-axis, three-value ledger can form when it splits its budget into two non-interfering draws.

One further point closes the seam before a reviewer can open it: why the two draws compose *quadratically*, as a sum of squares, rather than linearly as v + s = c. The quadratic composition is the signature of simultaneous independence. Two draws that run one after the other compose linearly, their costs adding along a single line; two draws that must both be satisfied *at the same tick*, neither pausing for the other, compose as a sum of squares, because simultaneous independent constraints combine as orthogonal components, not as collinear ones. Translation and the temporal scan run together every tick — the configuration does not stop moving to hold its tilt, nor stop holding its tilt to move — so their composition is quadratic by the fact of their simultaneity. The form is fixed by the simultaneity, not assumed to fit the Lorentz factor; the same simultaneity that forces the right angle forces the squares.

Let the spatial translation rate be v — the share of the ninety-seven percent spent stepping through space. Let the temporal scan retain a rate s — the share held back to track the tilt. Because the two are orthogonal legs of a triangle with hypotenuse c:

$$v^2 + s^2 = c^2$$

This is not an assumption; it is the sum rule of Section 3 read geometrically, with the zero-cross-talk independence of "stepping through" versus "holding against" supplying the right angle. Solving for the internal scan rate that remains at translation speed v:

$$s = c \cdot \sqrt{(1 - v^2/c^2)}$$

The internal scan rate retained by a configuration moving at v is the baseline rate c reduced by exactly the factor $\sqrt{(1 - v^2/c^2)}$. This is the rate at which the configuration can still hold its tilt and run its temporal scan while translating at v. The Lorentz factor therefore emerges not as an empirical convenience but as the only way a three-axis ledger can divide a fixed budget between two non-interfering draws without leaving the account unclosed.

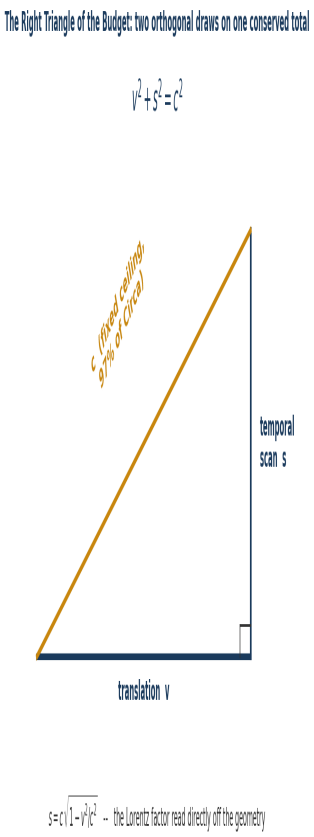


Figure 3 - The Right Triangle of the Budget. Translation v and the temporal scan s are orthogonal draws on one conserved total, the fixed ceiling c. The Pythagorean constraint and the Lorentz factor are read directly off the geometry.

5.2 Reading the Factor and the Unreachability of c

At v = 0 the factor is 1: a configuration at rest — in the active, balanced sense of Paper 6, never truly still — retains its full scan rate and holds its tilt with the whole non-translation budget. As v rises toward c the factor falls toward zero: more and more of the circa is consumed by translation, and the scan rate left to hold the tilt shrinks. The internal clock of the configuration — which *is* the rate of its temporal scan — slows by exactly this factor, which is the time-dilation Paper 13 develops; here it is the mass law's twin, the same factor read on the scan rather than on the workload. The standing workload required to *maintain* the tilt against a shrinking available scan rate rises as the reciprocal of the factor, $1/\sqrt{(1 - v^2/c^2)}$: this is the escalation of ledger workload Paper 7 named, the reason a configuration grows harder to accelerate as it speeds up. The effective mass — the workload of re-cutting the tilt — rises without bound as v approaches c, because the factor in the denominator approaches zero.

The derivation makes the limit a hardware fact, not a postulate. To reach v = c the factor $\sqrt{(1 - v^2/c^2)}$ must reach zero, which means the internal scan rate s must reach zero: the configuration would have to commit to translation every part of the budget the identity floor does not claim, and retain nothing to run the temporal scan that holds its tilt. But a tilt that cannot be scanned cannot be held, and an unheld crease dissolves back into the baseline. A massive configuration therefore cannot reach c, not because of an imposed speed limit, but because c is the velocity at which it would cease to be a massive configuration at all — it would have spent the last of the budget that held it in existence. The photon reaches c only because it never had a tilt to hold: it runs the identity scan alone on the three-percent floor, needs no temporal scan, and so loses nothing beyond that floor to the holding of a tilt, committing its full remaining ninety-seven percent to translation. The unreachable limit for mass and the natural home of the photon are the same point on the sum rule, approached from the two sides of having a tilt and having

none.

5.3 The Isotropic Extension of the Pythagorean Constraint

The budget split was derived above on a single line of motion, with v the translation rate along that line. The medium is three-dimensional, so the rule must hold when a configuration translates diagonally across all three native axes at once, and it does, without modification to its logic. A diagonal translation is the vector sum of steps across the three axes, so the total translation rate is fixed by the three-axis Pythagorean sum of its components:

$$v^2 = v_x^2 + v_y^2 + v_z^2$$

Substituting this into the budget constraint of Section 5 gives the full three-axis form of the sum rule:

$$v_x^2 + v_y^2 + v_z^2 + s^2 = c^2$$

The internal scan rate s therefore depends only on the *total* translation rate v, never on its direction: a configuration moving at a given speed retains exactly the same scan rate whether it travels along one axis or diagonally across all three, because only the magnitude of the vector sum enters the constraint. This makes s an isotropic scalar — it throttles uniformly with total speed and is blind to heading. Two consequences follow that the framework needs. First, the slowing of the temporal scan, and so the time-dilation that Paper 13 develops from it, is perfectly symmetric across the axes: it depends on how fast a configuration moves, not which way, which is the structural reason no orthogonal-isotropy test finds a preferred direction (the manifold-scale invariance of Paper 6). Second, because s is a single scalar shared across all three axes rather than three separate per-axis budgets, a configuration moving through three-dimensional space never deforms or tears its baseline alignment — the scan that holds its tilt together draws down evenly, leaving its internal geometry intact regardless of travel direction. The single ceiling c, split against an isotropic scan, is what lets a configuration move freely in three dimensions while holding one undistorted identity.

This single-scalar reading also closes a trap that cost the early relativistic-mass literature decades: the split into "longitudinal" and "transverse" mass, in which an object was said to resist acceleration differently along its direction of motion than across it. The trap arises only if one imagines separate, per-axis budgets, so that motion along X drains an X-budget and leaves a Y-budget fresh. The framework has no such per-axis budgets. The scan rate s is a single, configuration-wide clock rate, throttled globally by the total speed; there is no fresh sideways pool to draw on. When a configuration already translating fast along X is pushed along Y, the push does not meet a rested transverse mass — it meets a configuration whose entire processing capacity is already drawn down by its total speed, with the same reduced s available to re-cut its geometry in any direction. The resistance to a push is therefore identical from every heading, set by the one global scan rate, not by a direction-dependent mass. There is no longitudinal-versus-transverse distinction because there is one budget and one scalar, not three, and the configuration's geometry stays intact across all headings precisely because the throttle is global rather than per-axis.

5.4 The Escalation of the Resolution Path

The reciprocal growth of the maintenance workload, $1/\sqrt{(1 - v^2/c^2)}$, was read above as an algebraic fact about the budget. It has a direct mechanical meaning worth making explicit, because it is what the escalating workload physically *is*. The temporal scan does not merely hold a tilt in place; it tracks the configuration's route of resolution back toward the photon baseline (Section 2). As an accelerating force drives v toward c, two things happen at once, and they compound. The configuration drifts structurally further from State 9 — its departure deepens — so the route of resolution back to baseline becomes longer and more complex, demanding more computation to trace. At the very same time, by the budget rule, the scan rate s available to do that computing is shrinking toward zero. The workload therefore escalates from both ends simultaneously: a lengthening path to compute, and a starving budget to compute it with.

This is the mechanical content of the reciprocal factor, and it is why the escalation is so steep near c. Mass increase is not a pile-up of stuff, and it is not a buffer filling up; it is the sheer computational workload of a deep structural departure trying to find its resolution path on a processing budget that is running out of cycles. The configuration grows harder to accelerate because each further increment of speed lengthens the route home while leaving less capacity to calculate it — the temporal scan is being asked to solve a harder problem with fewer resources on every tick. At the limit the path to baseline would be longest and the budget to trace it would be zero, which is the same self-dissolution the previous paragraph reached by the algebra: the configuration cannot reach c because at c it can no longer compute its own way home, and a departure that cannot trace its route of resolution cannot be held.

6. What the Single Ruler Buys

Stated this way, mass stops being a primitive and becomes a reading, and several facts the standard account must postulate separately fall out of the one accounting.

6.1 Consequences That Fall Out of the One Accounting

Particle-antiparticle mass equality falls out automatically. The electron and positron are the same configuration tilted in opposite directions (Paper 4); the work to cut a tilt of a given magnitude is the same whichever direction it is tilted, so the standing ledger entry m is identical for both. The standard account postulates this equality through CPT symmetry; here it is a consequence of tilt-magnitude being direction-independent, requiring no separate principle.

Photon masslessness and its speed fall out automatically. The photon is the zero-tilt configuration; with no tilt there is no temporal scan, no budget diverted beyond the three-percent identity floor, no standing workload — no mass — and its full remaining ninety-seven percent is free for translation at c. The standard account assigns the photon zero mass by hand; here zero mass and speed c are the same fact, the fact of running the identity scan alone.

The mass hierarchy by complexity falls out as a prediction rather than a table of inputs. A configuration's mass is the workload of its manifold scan, which scales with the magnitude and complexity of its tilt: the single-axis first step of State 7 or State 8 is the lightest possible departure; the three-rotation climb to the cubic lock of State 1 is heavier; the dense, multi-junction tilts of composite matter are heaviest, their scan-load the deep bottleneck of Paper 7.5's spectrum. The standard account takes the lepton and quark masses as free input parameters; this framework requires them to order by configurational complexity, a constraint rather than a freedom.

The lightest departures order correctly, with no floor invented by hand. The ordering predicts that the smallest possible mass belongs to the smallest possible tilt — a single-axis first step that has only just left the baseline, holding the shallowest departure the medium permits above the resolution floor. Such a configuration runs only the lightest temporal scan and so carries only the smallest mass the framework allows, and the standard account's lightest particles are, on this reading, the shallowest tilts rather than fine-tuned parameters. The framework does not need to assign a separate small number to each light state; it requires only that shallower tilts carry less mass by the same rule that gives deeper tilts more, with the resolution floor of Paper 4 setting how shallow a tilt can be and still stand. The detailed classification of these shallowest configurations is the work of the later state papers; what this paper fixes is only that the lightest mass is the shallowest tilt, by the one rule.

None of these requires a new entity, a hand-tuned constant, or a second ruler. Each is the one accounting — tilt-magnitude against the photon baseline, held by the manifold scan, paid out of the fixed circa — read in a particular case.

6.2 The Quantum Threshold of the First Tilt

The ordering above runs from the lightest departure to the heaviest, and the lightest end of it deserves its exact mechanics, because it is where mass switches on and where the resolution floor of Paper 4 fixes the smallest mass

the framework permits. Mapping the scan budget across the first three rotations — the climb from baseline to the electron's cubic lock — makes the threshold precise.

State 9, the photon baseline. Pure zero at the central junction. The configuration pays the three-percent identity floor and, with no departure to track and so no temporal scan, commits its whole remaining ninety-seven percent to translation at c. Mass is exactly zero, and this is the absolute origin of the ruler, not a small mass rounded down.

State 7 or State 8, the first-tilt step. A single axis steps a non-zero value into the centre while the other two hold zero. This is the first departure the medium can register, and it is the moment the temporal scan switches on: a fraction of the circa, previously committed to translation, is diverted to run a primitive single-axis temporal scan that tracks the new, shallow route back to baseline. Because this is the *smallest* tilt the medium can resolve and still hold — one axis, one step, no smaller increment permitted below the Discrete-Step Resolution Floor of Paper 4 — the mass it carries is the **minimum mass quantum** of the framework: the least non-zero departure that can stand without dissolving. There is no mass between zero and this step, because there is no tilt between zero and this step; the floor is what makes the smallest mass a definite quantity rather than an arbitrarily small one.

State 1, the electron or positron. The three-rotation climb is completed, carrying the single-axis first step up to a saturated three-axis triplet lock — the cubic mark in which all three axes hold the same non-zero value. At this lock the identity scan and the temporal scan reach a stable geometric equilibrium: the departure is deep enough and symmetric enough to be self-sustaining, the manifold scan settling into a permanent standing pattern rather than a shallow, easily-undone step. This is why State 1 is a stable mass standard and the single-axis first tilt is not the electron: the electron is the *locked* end of the climb, three rotations deep, while State 7 and State 8 are its first rung. Mass rises monotonically along the climb — deeper into the rotations, heavier the standing scan — so the sequence State 9 to State 7/8 to State 1 is also the sequence zero mass, minimum-quantum mass, stable-standard mass, each step a further commitment of the circa from translation into the scan that holds the deepening tilt.

This threshold mapping is what lets the framework speak of a smallest mass at all. The standard account, lacking a resolution floor, has no smallest meaningful tilt and so no natural minimum mass; it can place a light particle's mass anywhere on a continuous open scale and must fix it by measurement. The framework fixes the floor of the scale by hardware: the minimum mass is the single-axis first tilt at the resolution floor, and every heavier mass is a deeper or more complex departure counted upward from it.

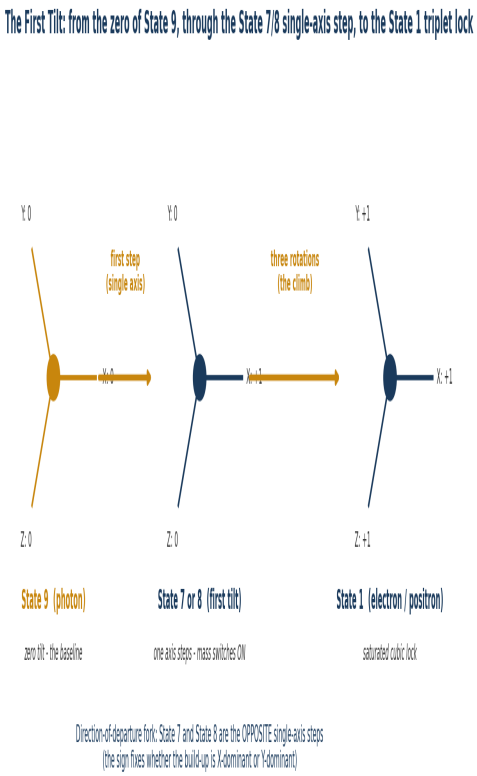


Figure 2 - The First Tilt. From the pure zero of State 9, a single axis steps to the State 7 / State 8 first mark (the direction-of-departure fork), and three rotations carry the climb to the saturated State 1 cubic lock. Mass switches on at the first step, not at the lock.

6.3 Why Complexity Is Slow: The Velocity Hierarchy from the Manifold Scan

The accounting of Section 2 yields an immediate proof of why simple configurations move fast and complex ones move slowly, drawn entirely from the cost of the manifold scan — the identity scan plus the temporal scan it triggers. The hierarchy runs from the photon, whose scans cost the least and which moves fastest, to the singularity, whose scans consume everything and which cannot move at all.

The photon — the smallest scan, no temporal scan. The photon is the minimum of structural complexity: it holds no tilt and has an elementary single-state identity, so it pays only the three-percent identity floor off Circa. With no tilt it runs no temporal scan at all, so the whole ninety-seven percent that remains goes to translation. That is why the photon moves at the full translation ceiling, v = c.

The electron — a heavier identity scan, and the temporal scan it triggers. An electron carries a stable cubic tilt, a more complex identity than the photon's, so its identity scan climbs above the three-percent floor — it has a full three-axis triplet lock to audit and confirm, not a bare zero. That heavier identity is the cause; the temporal scan is the effect it triggers, because a configuration that has just confirmed "I am not the baseline, I hold a tilt" must then compute how that tilt resolves back toward zero. Both scans draw from Circa, and both are larger than a photon's, so together they leave less of the ninety-seven percent for translation. The electron is therefore forced to translate below c — slower than the photon by exactly the budget its greater complexity costs it across both scans.

Complex atomic configurations — resource starvation. A complex atom is a dense network of multi-axis tilts and nested states, every one of which must be confirmed on every tick. Its identity scan is therefore enormous — a vast

web of junctions to audit just to hold the structure together — and that heavy identity triggers an equally heavy temporal scan, since tracking the resolution paths of that entire web requires a massive amount of sequential coordinate math. Both scans climb together and both draw from Circa, so the translation budget left over is tiny. This is the explicit mechanical proof of the model: heavy things move slowly because the overwhelming share of their budget is spent *existing* — confirming a complex structural identity and computing its resolution — leaving almost nothing for travel.

The singularity — the v = 0 limit. Carry the ratio to its extreme. As structural complexity and density climb, the identity scan and the temporal scan it triggers grow together until, at the limit, the two together consume the entire Circa: the configuration's identity is so dense and so multiply-tilted that confirming it, and computing the immense resolution it demands, swallows the whole budget before a single unit is left for translation. With nothing left, there is no v: the configuration's translation budget drops to absolute zero. It cannot move through space at all, because one hundred percent of its Circa is locked into knowing what it is and computing the route of resolution back to baseline. This is the singularity in the framework's reading — not an infinite point-mass with a divergent field, but a localized processing standstill, a configuration so complex that the cost of merely existing has consumed everything it would need to go anywhere. The velocity hierarchy is bounded at both ends by the budget: the photon at v = c, whose scans cost the least, and the singularity at v = 0, whose scans cost everything.

The vast velocity gap from a photon to a complex configuration is therefore not an incidental fact about light being special; it is the direct readout of how much of a fixed processing budget each one must spend simply being itself. Mass scaling and velocity limitation are the same accounting seen twice: structural complexity consumes the budget through the prior skim, and what the skim consumes is exactly what can no longer be spent on motion. The simpler the identity, the cheaper its existence and the more of c it keeps; the more complex the identity, the deeper the skim, and the more thoroughly the configuration is pinned in place — to a standstill, at the limit.

7. The Architectural Division of Labor

This paper establishes the foundational ledger for internal mass-energy: mass as the cost of the manifold scan, $E = m \cdot c^2$ as displacement cost, and the Lorentz factor and the unreachability of c as budget accounting. What it does not derive here, it does not leave open — it hands forward as a fixed boundary condition that the later papers are structurally constrained to obey. This is a division of labor, not an admission of incompleteness, and the distinction matters: because this ledger balances exactly on every tick, the quantities it fixes are not loose ends but mandatory inputs the rest of the architecture must honor.

The gravitational reading of mass is built in Paper 10. It is not a separate mechanism but the external face of the very departure this paper accounts internally; Paper 10's ambient baseline load and resolution-seeking drift are constrained to reproduce, from the outside, the same tilt-magnitude this paper fixes from the inside, because internal and external readings of one geometric fact cannot disagree. The time-dilation that the budget split produces is built in Paper 13: the slowing of the temporal scan by the factor $\sqrt{(1 - v^2/c^2)}$ derived here is not re-derived there but inherited, the mandatory clock behavior any configuration obeying this paper's sum rule must show. The composite mass of bonded matter is built in Paper 16: how the scan-loads of many tilted sub-configurations combine into the mass of a planet or a craft is constrained by this paper's rule that mass is scan-cost, so Paper 16's composition law must return, for a single configuration, exactly the mass this paper assigns. The detailed classification of the shallowest single-axis tilts is the work of the later state papers, bounded below by the minimum mass quantum fixed here. Each is a locked coordinate allocation: this paper sets the boundary conditions, and the named papers are obligated to satisfy them, not free to overwrite them.

What this paper commits to, and stakes itself on, is the single accounting: that mass is the running cost of holding a tilt away from the photon baseline, that energy and mass are that one tilt read two ways with c^2 the area of the processing boundary between them, and that the law of mass increase toward c is the budget split read as a right

triangle whose closed form is the Lorentz factor. Mass is expressed throughout as departure from the one baseline, measured against the one resolution floor, in the one ruler this series permits — never in kilograms, and never as a primitive.

8. The Failure of Continuous Formalisms

The accounting built above does more than derive mass; it dissolves a set of pathologies the standard account has learned to live with, and it is worth stating plainly what those pathologies are and why the ledger never incurs them. The standard account guards its structure with mathematical devices — mass diverging to infinity, unphysical singular points, unexplained squared constants — and has come to treat its own inability to close these expressions as features of nature rather than what they are: artifacts of describing a discrete processing medium with continuous mathematics. Four contrasts make the point.

The infinite-mass pathology. Standard relativity holds that as an object accelerates toward c its mass grows toward infinity, requiring unbounded energy to push further, with no mechanical account of *why* the resistance climbs. The ledger gives the mechanism and removes the infinity together. Circa is fixed; as translation v claims more of the ninety-seven percent, less remains for the temporal scan that holds the tilt, so the workload of holding the tilt against a starving budget climbs as the reciprocal factor derived in Section 5. The object does not gain infinite stuff; its internal accounting ratio shifts as it redlines a finite budget. It can never reach c because to do so it would have to drive its temporal scan to zero, and a configuration that cannot run its temporal scan cannot hold its tilt — it dissolves rather than arrives. The barrier is real and the mechanism is finite; the infinity was only ever a continuous formalism divided by a vanishing quantity.

The quadratic composition. Reviewers accustomed to sequential bookkeeping ask why the budget composes as $v^2 + s^2 = c^2$ rather than the linear v + s = c. The answer is simultaneity, established in Section 5.1: translation and the temporal scan run at the same tick, neither pausing for the other, and simultaneous independent draws compose as orthogonal components — a sum of squares — where sequential draws would compose collinearly, as a sum. The quadratic form is not an algebraic convenience adopted to reproduce known results; it is the mandatory signature of two constraints satisfied at once on one processor.

The squared constant. In the standard account c^2 enters $E = m \cdot c^2$ as an empirical conversion factor with no pictured shape — a number from the laboratory. In the ledger, c^2 is the geometric area of the processing boundary, derived in Section 4 and Appendix B as the product of two simultaneous orthogonal boundary sweeps each running at rate c. The universe multiplies the two because it is measuring an area, not summing two motors. The squared constant is a surface, not a coincidence.

The singular point. Where the standard account collapses a black hole to a point of zero volume and infinite density — dividing by zero and forfeiting predictability — the ledger reads the same object as a localized processing standstill, developed in Section 6.3 and Section 9 below: a junction whose manifold scan has claimed the whole of its Circa, driving translation to zero. Time does not stop there by a warping of geometry; it stops because the local processor has no cycle left to advance a clock. There is no division by zero, only a budget fully allocated — a clean discrete limit where the continuous account finds an infinity.

The through-line is one claim: the quadratic budget rule and the squared boundary area are not mathematical conveniences but the strict consequences of simultaneous, multi-axis updates executing across a discrete resolution floor. Replace the continuum with the ledger, and the infinities become finite allocations, the singularities become standstills, and the unexplained constants become geometric measures. The standard account's pathologies were never in the universe; they were in the choice to model a discrete medium as a smooth one.

9. Black Hole Growth as Ledger Saturation

The singular point deserves its own treatment, because the standard account carries an unresolved tension there that the ledger turns into a mechanism. General relativity collapses a black hole's interior to a point of zero volume, yet the same account has the event horizon grow in radius each time the hole absorbs matter. A zero-volume point that nonetheless occupies more space the more it swallows is a contradiction the standard account manages numerically — a larger value in the metric — without a physical picture of why nothing should expand. The ledger supplies the picture, and it follows directly from the velocity hierarchy of Section 6.3.

Read as processing, infalling matter is not crushed out of existence; it is a large influx of complex, heavily tilted configurations, each carrying a heavy manifold scan. Piled into one region, their scans drive the local processing load toward saturation: the medium there approaches the limit where the identity-and-temporal scan claims the whole of the available Circa, and translation falls to zero — the standstill of Section 6.3, reached now not by a single extreme configuration but by density. The event horizon, in this reading, is not a magical surface of no return but the boundary of the region that has been driven to processing standstill, where the local budget is fully consumed and no cycle remains to advance motion or a clock.

Growth follows from the accounting rule that the books must balance every tick. When still more matter arrives at an already-saturated junction, the medium cannot simply discard the load — that would violate the generative accounting held since Paper 1 — and it cannot process it locally, because the local Circa is spent. The one remaining move is to extend the standstill outward, drawing further neighbouring regions into saturation to carry the overflow. The horizon's physical growth is the visible edge of that spreading saturation: the region of budget-exhausted medium widens because the processing debt of the infalling matter must be absorbed somewhere, and the only somewhere available is the adjacent medium. The hole grows not because a point of nothing is gaining weight, but because the zone of local processing standstill is expanding to balance a load the interior can no longer carry. This ties the framework's most extreme object to the identical budget rule that governs a photon and a drifting mass, and it is offered here as the qualitative mechanism; the coordinate detail of the saturation front is a declared downstream allocation to the gravitational machinery of Paper 10.

10. Conclusion

This paper set out to answer the question mass actually asks — not how much of a substance a body contains, but what a configuration must continuously *do* to stand as a departure from the photon baseline — and the answer is one accounting rule read from end to end. Every configuration is allotted the same total budget, the Circa. Existence is paid for first: a fixed three-percent identity skim, taken off the top because energy cannot come from nothing, the framework's own three appearing in the budget as it appears in the axes, the values, the triangle, and the rotations. What remains — ninety-seven percent of Circa — is the fixed universal translation ceiling, the constant every observer records as c, the same for a photon and for a planet. The photon reaches that ceiling because it holds no tilt and so runs no temporal scan: massless means the temporal scan is exactly zero, not that existence is free, and light's zero proper time falls out of the same fact. The first single-axis step of State 7 or State 8 switches the temporal scan on, and from that moment the identity scan and the temporal scan grow together — the one confirming what the configuration is, the other computing how it resolves back to zero — both drawing from the one budget, both stealing from motion. Mass is that theft, stated mechanically and measured against the single ruler.

From that one rule the paper derived what the standard account postulates. The equivalence $E = m \cdot c^2$ emerged as topological displacement cost, with c^2 not an empirical conversion token but the geometric area of the processing boundary — two orthogonal sweeps, identity and its temporal mirror, executed simultaneously on every tick, their measures multiplying into an area precisely because they cannot run in sequence. The Lorentz factor emerged as the closed form of the budget split: translation and the temporal scan as orthogonal, zero-cross-talk draws on the fixed ceiling, composing quadratically because they are simultaneous, yielding $v^2 + s^2 = c^2$ and with it the exact factor $\sqrt{(1 - v^2/c^2)}$ that governs the slowing scan and the escalating workload. The unreachability of c became a

hardware fact — a massive configuration arriving at c would have shut down the very scan that holds it in existence — and the isotropy of the scan rate dissolved the old longitudinal-versus-transverse trap, because one global scalar throttles the whole configuration regardless of heading. The velocity hierarchy followed from the same books: the photon fastest because its scans cost least, complex matter slow because its scans cost most, and the singularity at the far end as a localized processing standstill — v = 0, the budget fully consumed by existence — with black hole growth read as that saturation spreading outward to balance a load the interior can no longer carry. Where continuous formalisms meet infinities, singular points, and unexplained squared constants, the discrete ledger meets finite allocations, standstills, and a boundary area.

The discipline of the series was held throughout, and it is worth closing on it because it is the point of the series. No mass was expressed in kilograms, and no dimensioned human unit was smuggled in as a constant: every result is a dimensionless ratio of one budget, invariant under any choice of translation unit, with the resolution floor a-k entering only as the non-zero logical delimiter that makes a ledger entry possible — never as a scalable metric parameter. No new entity was invented to bridge a gap. The composite mass of bonded matter, the gravitational reading of the same departure, the time-dilation carried by the slowing scan, and the coordinate detail of the saturation front were not left open; they were locked forward as declared boundary conditions to Papers 16, 10, and 13, each obligated to reproduce what this paper fixes. What Paper 8 stakes itself on is the single claim beneath all of it: that mass is not a token assigned to matter but the running cost of holding a departure from the photon baseline, paid every tick out of one budget, on one ledger, against one ruler — and that when the books are kept that strictly, the equations the standard account treats as mysteries arrive as line items.

Appendix A: The Discrete-Step Resolution Floor and the Minimum Mass Quantum

Section 4 (Step 3) and Section 6.2 both rest on the Discrete-Step Resolution Floor first established in Paper 4 (Section 3 and Appendix). This appendix lays out the mechanics of that floor explicitly: how a single-axis step is measured at the resolution boundary, and why any tilt shallower than one step dissolves back into the State 9 baseline rather than standing as a small mass.

The floor as the medium's minimum interval. The medium does not resolve displacement continuously. It has a smallest interval below which no crease can be registered as a standing feature — the resolution floor, written algebraically as a-k, with the single load-bearing property that it is strictly non-zero (a-k ≠ 0). The framework carries a-k as an algebraic entity only; it assigns it no decimal value, because fixing a number would smuggle an external, hand-tuned marker into a framework whose only permitted reference is the photon baseline (Paper 4.5, Section 5). What matters for every result in this paper is not the size of a-k but the fact that it is non-zero.

Why the floor must be non-zero. Consider a single axis carrying a displacement of magnitude d away from its zero. The local gradient the medium must resolve across the boundary scales as the displacement divided by the interval over which it is read. If that interval could shrink to zero — if a-k were zero — the gradient for any non-zero d would diverge, and an injected displacement would produce an instantaneous, infinite gradient across the continuous boundary. A medium cannot hold an infinite gradient; it would tear. The non-zero floor is therefore not an optional parameter but a structural necessity: a-k ≠ 0 is the cushion that lets a displacement be held as a finite, standing gradient rather than an infinite, tearing one. This is the same circuit-breaker role a discrete floor plays against the infinities that force renormalization in the continuum account, here stated as the condition that lets a tilt stand at all.

The single-axis step at the boundary. A first tilt is one axis stepping from 0 to its first resolvable non-zero value. By the floor, the smallest such step is one unit of a-k: the axis cannot hold any value between 0 and a-k, because there is no resolvable interval there for a gradient to stand in. The displacement of the first tilt is therefore exactly one floor-unit, and the standing gradient it presents — the thing the temporal scan must hold — is that one-unit displacement read across the boundary. This one-unit standing gradient is the minimum non-zero workload the manifold scan can carry, and the mass it corresponds to is the **minimum mass quantum**: the cost of holding the smallest gradient the medium permits.

Why a fractional tilt dissolves. Suppose an interaction delivers a displacement smaller than one floor-unit — a fractional tilt, d < a-k. The medium has no resolvable interval to seat it: there is no coordinate step shorter than a-k for the partial displacement to occupy as a standing gradient. With nowhere to stand, the fractional displacement is not held; on the next tick the boundary reconciles it back to the nearest resolvable value, which is 0. The crease relaxes to baseline. Mechanically, a fractional tilt is not a small mass — it is no mass, because it never becomes a standing gradient at all; it is reabsorbed into State 9 before the temporal scan can lock onto it. This is the precise sense in which mass switches on *at* the first floor-unit step and not below it: there is no continuous ramp from zero, because the floor forbids any standing displacement between 0 and one unit. The minimum mass quantum is a genuine quantum — the smallest step, with nothing beneath it but baseline.

This is why the mass scale has a definite floor rather than an open continuum. The standard account, lacking a resolution floor, has no smallest standing displacement and so no natural minimum mass, which is one of the structural failures audited in Paper 8.5. The framework's floor fixes the bottom of the scale as one a-k step at State 7 or State 8, with the photon's exact zero below it and every heavier mass counted upward from it.

Appendix B: The 720-Degree Double Traversal and the c-Squared Boundary Area

Section 4 (Step 4) derives $E = m \cdot c^2$ by identifying c^2 as the geometric area of the processing boundary across which a standing loop must converse, squared because the manifold scan is a double-sweep. This appendix charts that double-sweep explicitly: how the identity scan and the temporal scan operate across the boundary over a 720-degree cycle, and why their simultaneous operation forces the product $c \times c$.

Why a tilted configuration requires 720 degrees. A configuration at the photon baseline returns to its own orientation under a single 360-degree traversal, because it carries no tilt to track: its identity is the same at every point of the turn. A tilted configuration does not. Established in Paper 2, a configuration carrying a standing tilt must traverse its cycle *twice* — 720 degrees — to return to its original orientation, because the tilt couples the configuration's identity to its orientation in the medium, and a single 360-degree turn returns the orientation but leaves the identity inverted relative to its start. Only the second traversal restores both together. The two traversals are not redundant; they are two distinct sweeps the configuration must complete on every cycle to remain itself.

The two sweeps are the two scans. These two traversals are exactly the two scans of the manifold scan (Section 2). The first sweep is the identity scan: the traversal that confirms *what* the configuration is, carried over one 360-degree turn. The second sweep is the temporal scan: the traversal that tracks *where* the configuration stands relative to its route home, carried over the second 360-degree turn. The full 720-degree cycle is the union of the two — the manifold scan completing once. A tilted configuration runs both sweeps on every cycle; a photon, with no tilt, completes its return in one 360-degree sweep and runs the identity sweep alone.

Why the sweeps operate across a boundary, each scaling with c. Each sweep is not an internal abstraction; it is a conversation the standing loop holds with its neighbouring shapes across its processing boundary, and the medium processes every such conversation at its one rate, c (Section 2). The identity sweep traverses the boundary at rate c; the temporal sweep traverses the boundary at rate c. Each sweep, taken alone, carries one factor of c — it is a one-dimensional traversal of the boundary at the medium's rate.

Why the two together force a product, $c \times c$. The standing loop must satisfy *both* sweeps at once. It is not enough to complete the identity sweep and then the temporal sweep in sequence and add their costs; the configuration must hold its tilt against both sweeps simultaneously across the whole cycle, because the tilt that the temporal sweep tracks is the same tilt the identity sweep must keep confirming. Holding against two simultaneous traversals of a boundary is holding against the *area* they sweep out together — the product of the two traversal rates, not their sum. One sweep contributes its factor of c along one direction of the boundary; the second sweep contributes its factor of c along the orthogonal direction of the same boundary; and because both must be satisfied at once, the measure of what the loop holds against is their product, $c \times c = c^2$. The square is therefore the exact signature of the double traversal: two boundary sweeps, each at rate c, operating simultaneously rather than sequentially, so their measures multiply into the area of the processing boundary. This is the c^2 that converts a held tilt-magnitude m into the energy $E = m \cdot c^2$ required to hold it — not an algebraic scaling constant, but the two-dimensional measure of a boundary that must be swept twice at once on every 720-degree cycle.

The sequential-versus-simultaneous distinction made explicit. Were the two sweeps merely sequential — first identity, then temporal, one after the other — their costs would add, and the conversion factor would carry a single power of c. The factor is squared precisely because they are not sequential but simultaneous, and the reason they are simultaneous is that the temporal sweep is a *mirror* of the identity sweep, not an independent second computation. The identity sweep confirms what the configuration is; the temporal sweep reflects that same identity into the question of how it returns to baseline. Because the second is a reflection of the first, it cannot run later or elsewhere — a mirror image exists only at the same instant and on the same surface as what it reflects. The configuration cannot pause its identity scan to run its temporal scan, since a tilt left unconfirmed for even one tick dissolves (Section 2); both run every tick, across the same boundary, along orthogonal facets, identity along one and its mirror along the other. The measure of a value projected against its own simultaneous mirror is the product of the two, an area, not their sum. The 720-degree geometry makes the double traversal mandatory, the mirror relation

makes it simultaneous, and the two together make the conversion factor an area rather than a length. This is the topological content beneath the familiar c^2: a hardware processing constraint — confirm the identity and mirror it to baseline, at once, across a two-faceted boundary, on every cycle — standing where the standard account places an unexplained kinematic square.

Appendix C: Sample Ledger State Configurations

Section 6 traces the threshold where mass switches on from State 9, through States 7 and 8, to the cubic lock of State 1. The main text describes these as single-axis steps and three-rotation triplet locks; this appendix shows the literal central-junction signatures, in keeping with the standard of complete and transparent accounting fixed in Paper 1. Each configuration is written as its central-junction signature across the three native axes (X, Y, Z), with each axis holding one of three resolved values: +1, 0, or –1, in units of the resolution floor a-k of Appendix A.

State 9 — the photon baseline.

Configuration	X	Y	Z	Tilt magnitude	Scan	Mass
State 9 (photon)	0	0	0	0	identity only	0

Pure zero across all three native axes. No axis is stepped into the centre; there is no displacement to hold and no temporal scan, so the configuration pays only the three-percent identity floor and commits the remaining ninety-seven percent to translation at c. This is the absolute zero of the ruler.

States 7 and 8 — the single-axis first tilt. A single axis steps to one floor-unit while the other two hold zero. Which axis steps, and in which direction, is the direction-of-departure fork that fixes whether the configuration builds X-dominant or Y-dominant as further rotations are added (Section 2).

Configuration	X	Y	Z	Tilt magnitude	Scan	Mass
State 7 (X-dominant first step)	+1	0	0	one floor-unit	identity + primitive temporal	minimum quantum
State 7 (direction variant)	0	+1	0	one floor-unit	identity + primitive temporal	minimum quantum
State 8 (Y-dominant first step)	–1	0	0	one floor-unit	identity + primitive temporal	minimum quantum
State 8 (direction variant)	0	–1	0	one floor-unit	identity + primitive temporal	minimum quantum

Exactly one axis carries a non-zero value of magnitude one floor-unit; the others remain at baseline. This is the shallowest standing tilt the medium permits (Appendix A), so each carries the minimum mass quantum. State 7 and State 8 are the opposite single-axis steps: the sign of the stepped axis fixes the direction of departure, and that direction is what later determines the dominance of the built-up configuration.

State 1 — the saturated cubic lock (electron / positron). The three-rotation climb completes when all three axes hold the same non-zero value, a fully saturated and symmetric triplet.

Configuration	X	Y	Z	Tilt magnitude	Scan	Mass
State 1 (electron)	–1	–1	–1	three floor-units, saturated	identity + temporal at equilibrium	stable standard

Configuration	X	Y	Z	Tilt magnitude	Scan	Mass
State 1 (positron)	+1	+1	+1	three floor-units, saturated	identity + temporal at equilibrium	stable standard

All three axes are stepped to the same non-zero value, the saturated cubic mark. The electron and positron are the two directions of this triplet lock — (–1, –1, –1) and (+1, +1, +1) — identical in tilt magnitude and therefore identical in mass, the direction being the only difference between them (Section 6.1, particle-antiparticle equality). At this lock the identity and temporal scans reach a stable equilibrium, freezing the departure into a self-sustaining standard. The electron is the locked end of the three-rotation climb, three floor-units deep; State 7 and State 8 are its first rung, one floor-unit deep. The mass ordering — zero at State 9, minimum quantum at State 7/8, stable standard at State 1 — is the climb read as a ledger, each rotation a further commitment of the circa from translation into the deepening scan.

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APPENDIX D: VALIDATION REGISTER FOR THE MASS-AS-LEDGER CLAIM

(v2 addition, 2026-10-18. The original Paper 8 text is byte-untouched; this appendix is the only v2 change.)

D.1 The claim. Mass is not a substance a configuration contains but the running

cost of holding a tilt away from the photon baseline: the share of the fixed per-tick budget (Circa) diverted from translation into the manifold scan that holds the tilt. Stated in §2 ("Mass is the cost of the manifold scan that a tilt forces"); staked in §7 ("What this paper commits to, and stakes itself on, is the single accounting") and §10 ("What Paper 8 stakes itself on is the single claim beneath all of it: that mass is not a token assigned to matter but the running cost of holding a departure from the photon baseline").

D.2 Badge at source. [LEVEL A] — recorded here by author authorization,

Dervis Kadunic, 2026-10-18. The badge covers the mass-as-ledger thesis as derived in the prose proof named in D.3. The §2 statement of mass as manifold-scan cost is the definitional anchor of that proof and is disclosed as a definition; the equivalence E = m·c^2, the Lorentz factor, the unreachability of c, and the minimum mass quantum are derived from it and are covered by the badge.

D.3 Validation (prose section). The proof is the main-prose derivation,

recorded in the original text and unchanged by this register: §2 (the one budget and the two scans; the definitional statement of mass as manifold-scan cost); §4 Steps 1-6 (E = m·c^2 derived as topological displacement cost, with c^2 as the geometric area of the processing boundary); §5.1-5.2 (the Pythagorean budget split v^2 + s^2 = c^2, the Lorentz factor, the unreachability of c as a hardware fact, the isotropic extension); §6.2 (the minimum mass quantum at the State 7/8 first tilt); Appendices A-C (the resolution floor, the 720-degree double traversal, the sample ledger-state signatures); §7 and §10 (the paper's committed boundary conditions and what it stakes itself on). Paper 8 carries no machine check and no script registry for this claim; the validation type, in Paper 11.5 §0.5.6 terms, is "prose section" — no machine check exists, and none is claimed.

D.4 Scope. This register adds no physics, derives nothing new, and alters no

argument. It exists so that inheriting papers can cite a badge-at-source and a named validation record for the mass-as-ledger claim. The date of record is 2026-10-18.